



Instructor: Date:

Learner: Regional Branch:

Program: Center:

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Let's Start!

Understanding the World of Language Semantics



Learning Outcome:

Understand key information about the concepts, principles, and methodologies in simple spoken and written situations. Talk about programming techniques, use cases, kinds, and sequences in simple spoken and written situations.

Can Do Statements:

- I can understand the concept of language semantics.
- I can exchange information about language semantics.
- I can identify key features related to language semantics.

Ask your partner:

- How does a program execute in a computer?
- Which programming languages do you know?





Let's Explore!

Can you give me a hand* with this task?

*"Give a hand" is an expression that means to help.

```
String sql = getStatement();
resultSet = "select * from store";
if (resultSet = statement.executeQuery()) {
    result = true;
    setId(resultSet.getInt("storeId"),
    storeDescription = resultSet.getString("description"),
    storeTypeId = resultSet.getInt("typeId"),
    storeAddress = resultSet.getString("address");
}
```

Instructions: Read the text below.

Camila: Hi, what did you do yesterday?

Ruben: Hi! I wrote a report about language semantics, but I didn't complete it. I need some help with that task now because that topic is very complex. Can you help me in this moment?

Camila: Oh! Darling I'm looking for a new laptop that I need for my job, hum, but what do you need?

Ruben: Well, some specific information about Language Semantics.

Camila: Ok, we know that semantics is the field concerned with mathematical structure and the meaning of programming languages. It evaluates the meaning of strings syntactically defined in the specific programming language, depending on the needs of the users.

Ruben: I see, and do you have more information about it?

Camila: Remember, in the past, we used the definition of semantics for linguistics only, nowadays, semantics is an important part of computer science.

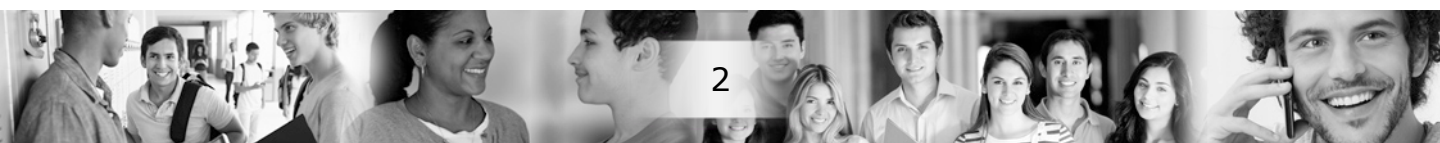
Ruben: You are right, semantics does the processes of a computer, I mean, it executes a program in a specific language.

Camila: Correct! At the beginning, programming languages were not able to solve complex problems. With its evolution, nowadays, they are successful, as they model some essential aspects of how to construct complex programs.

Ruben: So, is there any technique to understand semantics more easily?

Camila: Not at all. You have to study this topic more deeply to understand better. For example, there are four widely-used approaches to language semantics. These are: operational, axiomatic, denotational and logical semantics. However, many programmers dedicate their time to extend the language for programming at the operational level since they can execute those languages on a computer. Is it clear for you?

Ruben: I think so. Well, let me take some notes first, and then we go to buy your stuff.





Let's Practice!

Can you define Semantics?

Instructions: Match column A with column B.

COLUMN A

1. I wrote a report about language semantics, but
2. Language semantics
3. Semantics

COLUMN B

- a. Studies the meaning of programming languages.
- b. I didn't complete it.
- c. Evaluates the meaning of syntactical strings defined in the specific programming language.



Let's Practice!

I have some doubts* about this topic.

*When you have a "doubt" is because you are not sure about something.

Instructions: Select the correct answer in order to complete the sentence.

1. What did you _____ yesterday? I _____ my report about language semantics.
 - a. does, write
 - b. do, write
 - c. does, wrote
 - d. do, wrote
2. I _____ some doubts about the topic. Can you _____ me, please?
 - a. have, help
 - b. has. Helps
 - c. had, helping
 - d. having, to help
3. Do you _____ information about language semantics? Yes, I _____ for information on the net yesterday.
 - a. has, look
 - b. have, looked
 - c. have, look
 - d. has, looked



Extension Activity

Let's get this clear!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search for information about language semantics in software programming.
- Make a mind-map to show what you understood about language semantics.
- Share your mind-map in your class and discuss about the topic.
- Use the grammar points covered in this unit.

Self-Assessment

- I search for information about semantics in software programming.
- I express information through a mind map.
- I share information with other people.

Yes

No

☐ ☐
☐ ☐
☐ ☐


Target Words

- **string** (noun): a list of character codes
e.g. It evaluates the meaning of strings syntactically defined in the specific programming language, depending on the needs of the users.
- **task** (noun): something that you have to do because it is part of your job
e.g. I need some help with that task now because this topic is very complex.
- **syntactically** (adverb): related to the rules of syntax
e.g. It evaluates the meaning of strings syntactically defined in the specific programming language, depending on the needs of the users.
- **approach** (noun): method.
e.g. You have to study this topic more deeply to understand better. For example, there are four widely-used approaches to language semantics.
- **arbitrary** (noun): linguistics.
e.g. In the past, we used the definition of semantics for linguistics only, nowadays, semantics is an important part of computer science.



Learning Points

Simple Present and Simple Past with Time Expressions

You can use the **simple present** to refer to routines, daily events, and things that usually happen or actions that you frequently do.

You can use the **simple past** tense to talk about actions that started and finished in the past.

Simple Present

When you express general truths, you use **simple present**. For example, the sentence "Each country has a capital city" is a universal truth and it is stated in **simple present**.

Notice that for the subject pronouns **I, we, you** and the **simple present** form is equal to the infinitive, but for **she, he** or **it** you need to include *s/es* and you need the auxiliary **do/does** for the **negative** and **interrogative** form. There are some specific time expressions you can use for the **simple present**, for example, **every day, today, now, this year, nowadays**.

Example

*They design a new program **this year**.*

*They **don't design** a new program.*

***Do** they **design** a new program?*

*The company designs software **everyday**.*

*He doesn't design software **at the moment**.*

***Does Pedro** design software?*

Simple past

You use the **simple past** tense to talk about a complete event in the past. You can form the **simple past** tense by adding **ed, -ied** to the verb when it is regular. For irregular verbs, you have to learn its particular change. There are some specific time expressions you can use for the **simple past**, for example, **yesterday, last month, and ago**.

Example

*We designed a new program **yesterday**. (regular verb)*

*They studied in this institution **last year**. (regular verb)*

*He found new software 5 **years ago**. (irregular verb)*

You need auxiliary verb *did* for the **negative** and **interrogative** form, for all the subject pronouns.

Example

*We **didn't** design a new program.*

***Did** they study here?*

*He **didn't** find the program.*

Notice that when you use the auxiliary **did**, the verb takes the infinitive form.

Look at the chart of **simple past**:



Present Simple			
	Affirmative	Negative	Questions
I/you/we/they	I find a new program. You send an email.	We don't send new products.	Do they get products?
He/she/it	He sells software	She doesn't buy software.	Does she buy a computer?

Simple Past			
	Affirmative	Negative	Questions
Regular verbs I/you/we/ they/He/she/it	I watched a new program.	We didn't send new products.	Did they get products?
Irregular verbs I/you/we/ they/He/she/it	He sold software	She didn't buy software.	Did she buy a computer?

Examples

- *What did you do yesterday?*
- *I wrote a report about language semantics but I didn't complete it.*
- *We know that semantics is the field concerned with mathematical structure and the meaning of programming languages.*



Learning Points

Actions Verbs to Talk About Language Semantics

Action verbs are verbs that describe what the subject does.

The **action verbs** show in a sentence what the subject is doing physically or mentally.

It is important to consider that in English there are state verbs which are not commonly used in the **present continuous** because they describe a state, not specifically an action. On the other hand, the verbs that you can use in the **present simple** or **continuous** are the action or dynamic verbs.

Here some example of action

verbs: **define, execute, achieve, write, scale, show, and interpret.**

